Lesson 7
Data Handling Functions

Excel in Business

Five Important Functions in Excel

- INDEX
- MATCH
- OFFSET
- VLOOKUP
- INDIRECT

INDEX Function: Worksheet Index

Allows you to extract any value from a range by knowing its row and column.

	Delhi	Lucknow	Mumbai	Nagpur	Kolkata	Jaipur	Chennai	Bangalore
Delhi	0	983	1815	1991	3036	1539	2664	2612
Lucknow	983	0	1205	1050	2112	1390	1729	2052
Mumbai	1815	1205	0	801	1425	1332	1027	2404
Nagpur	1991	1050	801	0	1174	2100	836	1373
Kolkatta	3036	2112	1425	1174	0	2757	398	1909
Jaipur	1539	1390	1332	2100	2757	0	2359	3389
Chennai	2664	1729	1027	836	398	2359	0	1482
Bangalore	2612	2052	2404	1373	1909	3389	1482	0

INDEX Function:Worksheet Index(cont..)

What is the distance traveled if we undertake the round trip:

Bangalore -> Chennai -> Kolkata -> Nagpur -> Mumbai -> Lucknow -> Bangalore ?

INDEX Function:Worksheet Index(cont..)

The distance between Lucknow and Mumbai can be extracted by

=INDEX(D5:K12, 2, 3)

		1	2	3
		Delhi	**Lucknow**	**Mumbai**
1	**Delhi**	0	983	1815
2	**Lucknow**	983	0	1205
3	**Mumbai**	1815	1205	0

MATCH Function

MATCH function returns the relative position of an item in an array that matches a specified value in a specified order.

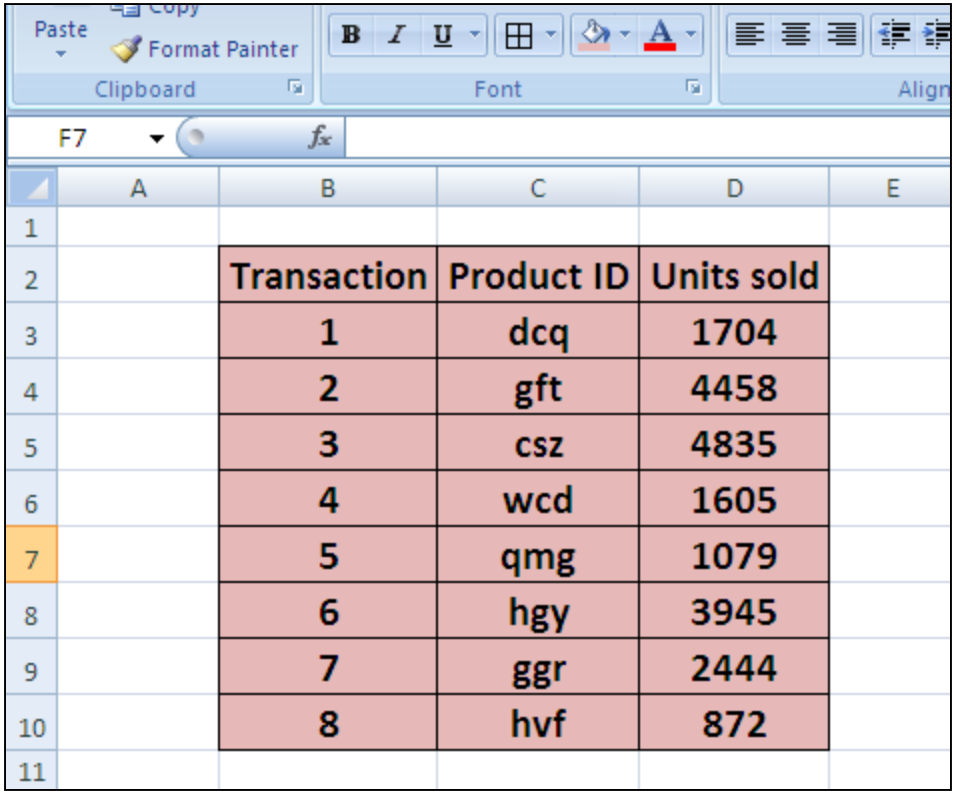
MATCH Function (cont..)

The MATCH function accepts three arguments:

- what are you looking for?
- where to look?
- how to look?

The 'how to look?' parameter signifies the type of search – loose or exact.

MATCH Function: example



Transaction	Product ID	Units sold
1	dcq	1704
2	gft	4458
3	csz	4835
4	wcd	1605
5	qmg	1079
6	hgy	3945
7	ggr	2444
8	hvf	872

If we want to know the row where max number of units occur, then the syntax is:

=MATCH(MAX(D3:D10), D3:D10,0)

The OFFSET Function

OFFSET creates a reference to a range that is a specified number of rows and columns away from a fixed cell.

OFFSET: Syntax

= OFFSET(reference, rows, cols, height, width

e.g.

=OFFSET(A1,2,3,1,1) will give the contents stored in cell D3

Goto worksheet offset!

LOOKUP FUNCTIONS

Lookup functions are extremely useful in situations where data needs to be classified into groups.

An Example: Worksheet Vlookup1

- A commodity was priced at $98 during the period Jan-Apr 1999, $105 during the period May-Aug 1999 and $112 during Sep-Dec 1999.
- We have a spreadsheet that displays some dates.
- How do we write a formula that tells us the price on that day?

An Example: Worksheet Vlookup1

Date	Price
1-Feb-98	
5-May-99	
23-Jan-99	
12-Sep-99	
25-Dec-99	

The prices are to be determined on the basis of the following table:

PERIOD	PRICE
1-Jan-98	NOT KNOWN
1-Jan-99	$98
1-May-99	$105
1-Sep-99	$112

An Example: Worksheet Vlookup1

- The VLOOKUP function offers a quick and easy way out.

- It needs to be passed the following arguments:
  1. What do you want to search?
  2. Where to search?
  3. What result to give out?
  4. How to search?

An Example: Worksheet Vlookup1

- In cell B2 issue the command

  =VLOOKUP(A2,$D$2:$E$5,2,TRUE)

  to have Excel determine the interval to which the given date (A2) belongs.

  The third parameter tells VLOOKUP to return a value from the second column (prices) of the range $D$2:$E$5.

An Example: Worksheet Vlookup1

- The last argument is specified as
  - TRUE if we want Excel to determine the interval provided the boundary points are in ascending order
  - FALSE if we want Excel to perform an exact search

REMEMBER

- The query is always searched in the first column of the range.
- The third parameter specifies the column from where the output is to be picked.
- The values in the first column must be in ascending order.

Another Example: Worksheet Vlookup2

- Compute the cost of advertising if the tariff plan reads:

Ads bought	Price per ad
First Five	$12,000.00
Next Five	$11,000.00
Next Ten	$10,000.00
Subsequently	$9,000.00

Another Example: Vlookup 2 (cont..)

So if you buy 14 ads then your total cost will be

5*$12,000 +5*$11,000+4*$10,000 = $155,000

Ads bought	Price per ad
First Five	$12,000.00
Next Five	$11,000.00
Next Ten	$10,000.00
Subsequently	$9,000.00

Another Example: Vlookup 2 (cont..)

Let us set up the lookup range as

1	$12,000.00	
6	$11,000.00	$60,000.00
11	$10,000.00	$115,000.00
21	$9,000.00	$215,000.00

The third column shows the cumulative prices.

Another Example: Vlookup 2 (cont..)

Observe that if our lookup value is 14 then VLOOKUP yields

- 115,000 if the result is picked from Col. 3
- 10,000 if the result is picked from Col. 2
- 11 if the result is picked from Col. 1 itself

Another Example: Vlookup 2 (cont..)

Now

Col. 3 + (14 – Col. 1 + 1) * Col. 2

= $155,000

= Cost of advertising We employ precisely this logic to our spreadsheet.

INDIRECT function

This function allows you to manufacture references to cells and ranges.

This facility has profound applications

INDIRECT: Syntax

= INDIRECT(reference_text)

e.g. = INDIRECT("A12") will give the value stored in cell A12

Goto worksheet Validity!